



TD2: Stabilizers

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09/10/2025

1 Introduction

$$J(\theta) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & e^{i\theta} \\ 1 & -e^{i\theta} \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

Clifford operation U	Pauli input P	Pauli output $UPU^\dagger$
$CNOT_{12}$		
H		
S		

Exercise 1 (Introduction to stabilizers).

1. What are the result of the following products:

- $SXS^\dagger$
- $SZS^\dagger$
- $Cx_{12}I_1 \otimes X_2Cx_{12}$
- $Cx_{12}X_1 \otimes I_2Cx_{12}$
- $Cx_{12}I_1 \otimes Z_2Cx_{12}$
- $Cx_{12}Z_1 \otimes I_2Cx_{12}$
- HYH
- $SYS^\dagger$

2. Show that $J(0)J(\alpha) = P(\alpha)$ and using $J(\pi)$ and $J(0)$ show that $X = HJ(\pi)H$

Exercise 2 (Manipulating stabiliser states).

1. What state is stabilised by the stabiliser group $S = \langle Z_1, Z_2, Z_3 \rangle$. We call this state $|\Psi_0\rangle$

2. What is the stabiliser group of the following states:

- $S_3Cz_{12}Cz_{23}H_2H_1|\Psi_0\rangle$
- $Cx_{13}Cx_{12}H_1|\Psi_0\rangle$

3. Write the following state in the canonical basis ($|000\rangle, |001\rangle, \dots$) and find its stabiliser group: $Cz_{12}Cz_{23}H_3H_2H_1|\Psi_0\rangle$

4. If you have a n-qubit state, how many terms is there at worst in the vector state representation? How many stabiliser generator is there at worst in the stabiliser formalism ?